

ORDER FULFILLMENT PERFORMANCE ANALYTICS

Technical Case Study

Portfolio Project 1 of 3 · Fran Morales · 2026

1. Project Overview

This project addresses a common operational challenge in supply chain management: the absence of centralized, decision-oriented visibility into order fulfillment performance. Without a unified analytical layer, service failures, profitability leaks, and structural inefficiencies remain buried in transactional data.

The objective was to build an end-to-end analytics solution that transforms raw supply chain data into actionable insights — demonstrating capability across data engineering, dimensional modeling, business intelligence, and statistical analysis.

Dataset

Source	DataCo Smart Supply Chain for Big Data Analysis (Kaggle)
Volume	180,519 order line items · 53 columns · 2015–2018
Grain	One row = one order line item
Format	Single denormalized CSV (Latin-1 encoding)

Core Business Questions

- What factors are associated with late deliveries?
- Where and under what conditions does service performance deteriorate?
- Which customers, products or segments combine high value and high risk?
- What operational patterns contribute to margin erosion or inefficiencies?

2. Technical Architecture

The project follows a structured analytical pipeline from raw data to portfolio deliverables:

Raw CSV → Python ingestion → PostgreSQL (raw schema) → SQL audit → PostgreSQL (analytics schema / star schema) → Power BI dashboard → Python EDA notebook

Technology Stack

Ingestion	Python (pandas, psycopg2, SQLAlchemy)
Storage	PostgreSQL — two schemas: supply_chain (raw) and analytics (warehouse)
Transformation	SQL (PostgreSQL) — dimensional modeling and ETL
BI Layer	Power BI Desktop — semantic model and Control Tower dashboard
EDA	Python — pandas, matplotlib, seaborn, statsmodels

3. Data Modeling

The raw denormalized table was transformed into a star schema dimensional warehouse consisting of one fact table and four dimension tables. Key design decisions are documented below.

Schema Design

- **Fact grain:** One row per order line item (order_item_id). Chosen because the source data is already at item level and all KPIs require item-level aggregation.
- **dim_date:** Synthetic calendar generated with generate_series, independent of source data. Integer primary key in YYYYMMDD format for join performance and human readability. Role-playing design: order_date_key and shipping_date_key both reference the same dimension.

- **dim_customer:** Natural key (order_customer_id). customer_region was dropped after data investigation revealed only US and Puerto Rico in the source data — regional analysis at customer level has no analytical value.
- **dim_product:** Natural key (order_item_cardprod_id). Two candidate IDs confirmed identical via query validation. Product Category and Department kept inside dim_product (MVP decision — no business question required standalone dimensions).
- **dim_shipping_fulfillment:** Surrogate key (SERIAL) — no natural unique identifier exists for a shipping profile. Grain: one row per unique combination of shipping mode, delivery status, market, region, country, state and city. 16,606 unique profiles loaded.

Key Technical Decisions

- **NUMERIC vs DOUBLE PRECISION:** Financial columns cast to NUMERIC during INSERT for exact precision. Confirmed during warehouse validation: a difference in the 12th decimal place between raw (DOUBLE PRECISION) and fact table (NUMERIC) — operationally irrelevant but evidence of correct design.
- **Correlated subquery for shipping_fulfillment_id:** Because the surrogate key was generated with SERIAL, a correlated subquery was required during fact table INSERT to look up the correct key by matching all seven shipping attributes.
- **Market attribute scope:** Investigation confirmed that Market is an order-level attribute, not a customer attribute — the same customer ID appeared with different Market values across 11,299 cases. Market was moved to dim_shipping_fulfillment.

Warehouse Validation

Three validation checks were run before connecting Power BI:

- **Row count:** fact_order_fulfillment: 180,519 rows — exact match with raw dataset.
- **Referential integrity:** LEFT JOIN checks on all four foreign keys returned 0 orphaned rows.
- **Business logic:** SUM(sales) comparison between raw and fact table confirmed match to 12 decimal places. Negligible floating-point difference explained by NUMERIC vs DOUBLE PRECISION precision.

4. Power BI Dashboard & KPIs

The Control Tower dashboard was built on the validated warehouse using Power BI Desktop connected to PostgreSQL via Parallels (Mac/Windows VM setup). The semantic model uses the star schema relationships directly, with DAX measures for calculated KPIs.

Dashboard Pages

- **Service Performance:** Late Delivery Rate (DAX: DIVIDE + FILTER), OTIF %, Average Order Cycle Time, trend analysis by year and quarter. Year and quarter slicers with cross-visual filtering.
- **Profitability:** Profit Margin % (SUM/SUM approach to avoid distortion from pre-calculated ratio column), Discount vs Profit Margin analysis using Power BI grouping on discount rate bins.
- **Risk:** Late delivery rate vs sales volume by customer segment and market using combo charts. Secondary axis fixed to 0–100% scale to prevent misleading visual distortion.
- **Executive Summary:** OTIF gauge with 70% target, average cycle time card, profit margin card, late delivery rate by shipping mode bar chart.

KPI Limitations

- **Shipping Cost Ratio:** Dataset contains no shipping cost column — only shipping time fields. KPI dropped and substituted with Discount vs Profit Margin analysis.
- **Perfect Order Rate:** Dataset captures only on-time delivery. Order completeness, damage, and documentation accuracy data are absent. KPI dropped to avoid presenting a duplicate of OTIF under a misleading name.

5. Python Exploratory Data Analysis

The EDA notebook extends the dashboard analysis by exploring patterns that aggregate KPIs cannot surface. The notebook is self-documenting with markdown cells explaining methodology and findings.

Analyses Performed

- **Negative profit investigation:** Row-level filtering revealed 33,784 loss-making orders (18.7%). Discount rate analysis showed near-identical mean discount between loss-making (10.2%) and full dataset (10.1%), with losses occurring even at zero discount.
- **Delivery time histogram:** Revealed a bimodal distribution hidden by the 3.50-day average: a dominant spike at 2 days (~56,000 items) and a flat plateau from 3–6 days (~29,000 items each).
- **Delivery time boxplot by shipping mode:** Confirmed First Class delivers in exactly 2 days with zero variation. Standard Class and Second Class distributions are statistically indistinguishable.
- **Correlation analysis:** Pearson correlation between order_item_discount_rate and order_item_profit_ratio: $r = -0.003$. Near-zero at the row level.
- **Correlation heatmap:** Full numeric correlation matrix surfaced: scheduled days vs late delivery risk (-0.37), real days vs late delivery risk ($+0.40$), quantity vs product price (-0.48).
- **OLS regression:** Multiple linear regression (discount rate, product price, quantity $\rightarrow$ profit ratio): $R^2 = 0.000$, F-statistic p-value = 0.559. All variables non-significant. Model explains nothing at row level.

Key Methodological Note

The near-zero row-level relationship between discount rate and profit ratio is not inconsistent with the Power BI finding of margin erosion by discount band. Grouping cancels out noise from other variables, surfacing a real but aggregate-level trend. Both findings are valid at their respective levels of analysis.

6. Key Findings

Service Performance

- **54.83% late delivery rate / 45.17% OTIF:** Structural operational failure affecting more than half of all orders.
- **First Class: 95.32% late delivery rate:** Premium tier is the worst performer due to a rigid 2-day promise with zero operational flexibility. Any deviation is flagged as late.
- **Standard Class and Second Class operationally identical:** Both spread from 2–6 days with median ~4 days. Four pricing tiers produce only two real delivery profiles.
- **Average cycle time 3.50 days is misleading:** Bimodal distribution: 2-day profile (First Class) and 3–6 day profile (all other modes) coexist in the same operation.

Profitability

- **Overall profit margin: 10.78%:** Aggregate figure within the 5–15% retail benchmark range.
- **18.7% of orders are loss-making:** Not visible in the aggregate margin. Root cause not identified — discount rate is not the driver.
- **Discount erodes margin in aggregate:** Clear decreasing trend from 0–5% to 15–20% discount bands. Not detectable at row level due to multiple interacting factors.

Risk

- **Risk is flat across segments and markets:** Late delivery rate varies by only 0.4 percentage points across Consumer, Corporate and Home Office segments. Same pattern across markets. The differentiator is sales volume, not risk.

7. Recommendations

- **Simplify the shipping tier structure:** Consolidate four modes into two tiers (express and standard) reflecting actual delivery capabilities. Reset customer expectations on First Class to reduce the structural late delivery rate driven by overpromising on a 2-day window.
- **Investigate loss-making orders:** 18.7% unprofitable orders require a cost-level investigation. Capturing shipping cost, return rates, and handling cost per order is a prerequisite for root cause analysis beyond what this dataset supports.
- **Invest in data completeness:** Shipping cost, order completeness, and damage data are missing. Without them, Shipping Cost Ratio and Perfect Order Rate cannot be calculated and the root cause of losses cannot be identified. A richer dataset would unlock significantly deeper analytical capability.

8. Limitations & Future Work

Data Limitations

- **Incomplete 2018 data:** Dataset covers 2015–2017 fully and Q1 2018 only. Year-over-year trend analysis is affected.
- **Missing cost data:** No shipping cost column. Shipping Cost Ratio and full cost-to-serve analysis are not possible.
- **Binary delivery flag only:** late_delivery_risk is a 0/1 flag. True Perfect Order Rate requires completeness, damage, and documentation accuracy dimensions.
- **Customer geography:** Customer Country contains only US and Puerto Rico — regional analysis at the customer level is not meaningful.

Future Work — Project 2

This warehouse is the clean data foundation for Project 2 (Predictive Operations Analytics). Planned extensions include:

- Predictive model for late delivery risk at order placement time
- Customer segmentation using clustering on purchasing behavior
- Time-series forecasting of fulfillment volume by market
- Integration of shipping cost data for full profitability analysis